Elements of Chemistry-II - CH1201

States of Matter (11 lectures): PD

Thermodynamics (11 Lectures): DC Sir

Chemical Kinetics (6 Lectures): PD/DC

Suggested Text Books:
(1) Physical Chemistry: Gilbert W. Castellan
(2) Physical Chemistry: P. C. Rakshit
(3) Physical Chemistry: Ira N Levine

Marks Distribution:	Tutorial Quiz:	30
	Mid-Sem Exam:	30
	End-Sem Exam:	40

PD Office: Polymer Research Center (between the ESS and VH)
Any day, preferably Saturday between 11:00 am -12:30 pm

Ideal Gas Equation:

A gas at equilibrium has definite values of pressure (P), volume (V), temperature (T) and composition (n). The state of a gas can be defined by these variables.

Boyel's Law (1662) : $V \propto \frac{1}{P}$ when n and T are fixed for the gas.

Charle's Law (1787) : $V \propto T$ " n and P " " " " ".

Avogadro's law (1811) : $V \propto n$ " P and T " " " " ".

when all variables are taken into account

$$V \propto \frac{1}{P} \times T \times n$$

or $PV = nRT \longrightarrow$ Ideal gas equation of state

This equn. is found to hold good most satisfactorily when $P \to 0$.
At ordinary T and P, this equn. is found to deviate about 5%.

Properties of Gases

$$PV = nRT$$

R = Universal gas constant. The value of R is calculated from the statement: At STP, one mole gas occupies 22.4 L.

Thus, $R = \frac{PV}{nT} = \frac{1 \text{ atm} \times 22.4 \text{ L}}{1 \text{ mol} \times 273 \text{ K}} = 0.082 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \text{K}^{-1}$

Also, $R = \frac{PV}{nT} = \frac{76 \times 13.56 \times 981 \text{ dyne cm}^{-2} \times 22.4 \times 10^3 \text{ cm}^3}{1 \text{ mol} \times 273 \text{ K}}$

$= 8.314 \times 10^7 \text{ erg mol}^{-1} \text{K}^{-1}$ **CGS Unit**

$= 8.314 \text{ J mol}^{-1} \text{K}^{-1}$ (since 1 Joule $= 10^7$ ergs) **SI Unit**

$= 1.987 \approx 2 \text{ Cal mol}^{-1} \text{K}^{-1}$ (1 calorie $= 4.184 \times 10^7$ ergs)

STP: Standard pressure and temperature, By IUPAC, T = 273.15 K and P = 0.987 atm
NTP: Normal pressure and temperature, By NIST, T = 293.15 K and P = 1 atm

IUAPC definition of STP: *273.15 K* and *1 atm* is discontinued since 1982.

Unit of P is dyne cm^{-2} (CGS)

$\text{Nm}^{-2} = \text{Pa}$ (SI)

1 torr = 1 mm of Hg

So, 1 atm = 760 torr

1 bar = 10^5 Pa = 750 torr

1 Pa = 10 dyne cm^{-2}

The moderate pressure is measured by manometer.

Also ; $PV = \left(\frac{w}{M}\right) \times RT$; w = wt. of the gas in gm

M = Molar mass of the gas

$$\Rightarrow P = \left(\frac{w}{V}\right) \times \frac{RT}{M}$$

$$= \left(\frac{d}{M}\right) RT \quad ; \quad \text{Density, } d = \frac{w}{V}$$

If the gas contains N number of molecules and N_A is the Avogadro number, then $n = \frac{N}{N_A}$

$$\therefore \quad PV = \left(\frac{N}{N_A}\right) \times RT$$

$$\Rightarrow P = \left(\frac{N}{V}\right)\left(\frac{R}{N_A}\right) \times T$$

$$= N' k T$$; N' = no. of molecules per unit vol. of the gas.

k = Boltzmann const.

$$= \frac{R}{N_A} = \frac{8.314 \times 10^{7}}{6.02 \times 10^{23}}$$

$$= 1.38 \times 10^{-16} \text{ erg} \cdot \text{molecule}^{-1} K^{-1}$$

Properties of Gases

P and T are intensive properties (independent of amount of the gas present in the container)

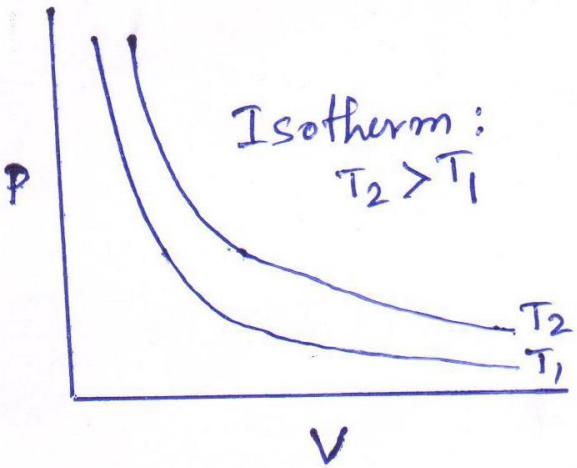
V is extensive property → dependent of the amount of the gas present in the container)

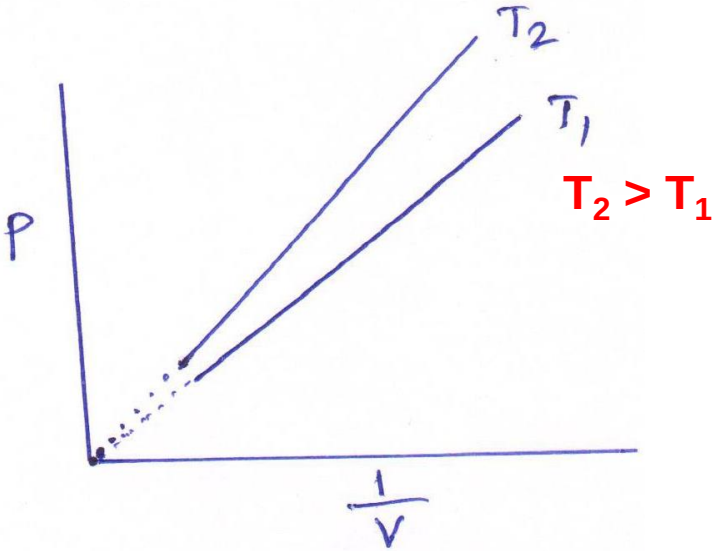
What about *n*, quantity of the gas?

Boyle's Law: $V \propto \frac{1}{P}$

$\Rightarrow PV = k$ (constant)

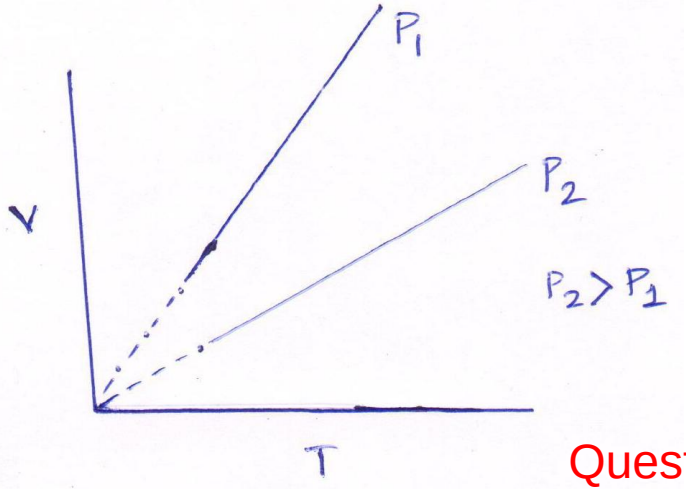
Plot : Hyperbola





$T_2 > T_1$

Charle's Law:



Isobars extrapolates to zero volume at T = 0 K or -273 ^oC

Question: Plot P *versus* T at $V_3 > V_2 > V_1$

Limiting Density:-

$$P = \left(\frac{d}{M}\right) RT \quad ; \quad d = \text{density of the gas in g/cc}$$

$$\Rightarrow M = \left(\frac{d}{P}\right) RT \quad ; \quad \frac{d}{P} = \text{Normal density}$$

$= \text{Density of the gas per unit pressure.}$

This equation is obeyed accurately by the gas when P is very low.

Thus, $M = \left(\frac{d}{P}\right)_{P \to 0} \times RT$; $\left(\frac{d}{P}\right)_{P \to 0}$ = Density per unit pr. at $P \to 0$

↑

Limiting Density of the gas at temp. T.

$M = \left(\frac{d}{P}\right)_{P\to 0} \times RT$ As $P \to 0$, $d \to 0$ and experimentally this low density cannot be measured accurately.

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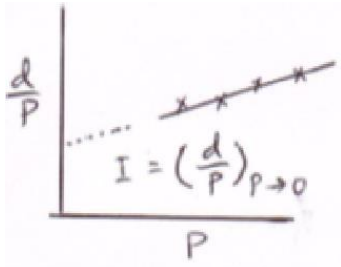

$M = \left(\frac{d}{P}\right) RT$

Thus, $M = I \times RT$

We can use graphical method

↓

Determine d of the gas at various pr., at low Pr.

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Plot $\frac{d}{P}$ vs. P → Extrapolate the st. line to get $\left(\frac{d}{P}\right)_{P\to 0}$

For a gas, M can be determined by limiting density method.

For O_2, $\left(\frac{d}{P}\right)_{P\to 0} = 1.4177$ at 25°C.

For the unknown gas, $\left(\frac{d}{P}\right)_{P\to 0} = 1.5177$ at 25°C

$$\text{Thus, } M_{unknown} = M_{O_2} \times \frac{\left(\frac{d}{P}\right)_{P\to 0} \text{ for unknown gas}}{\left(\frac{d}{P}\right)_{P\to 0} \text{ for } O_2}$$

$$= 32 \times \frac{1.5177}{1.4177} = 34 \text{ gm/mol}$$

The unknown gas is CH_3F

Properties of Gases: Abnormal Vapour Density

$$V.D = \frac{\text{Density of the gaseous state of the substance}}{\text{Density of } H_2 \text{ under same temp. and Pr.}}$$

$$d = \frac{PM}{RT}$$

Expected vapour density, $D_0 = \frac{M_0}{2}$

M_0 = Molecular formula weight of the substance.

For substances like NH_4Cl, PCl_5, N_2O_4, I_2, etc., it is found that vapour density (D) is observed to be less than that of the expected theoretical value (D_0). This is called abnormal vapour density.

Why?

Thermal dissociation:

$$NH_4Cl(g) \rightleftharpoons NH_3(g) + HCl(g)$$

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$$I_2(g) \rightleftharpoons 2I(g)$$

Due to dissociation, no. of moles of the system is increased. At constant T and P, the volume is proportionately increased. Since mass remaining same, density is decreased, resulting the decrease of vapor density.

Properties of Gases: Abnormal Vapour Density

The extent of dissociation(α), i.e., fraction of the total number of molecules dissociated can be related with the abnormal vapour density (D) of the substance at a given temp. and Pr.

consider a general representation of the thermal dissociation as

$$A(g) \rightleftharpoons nB(g)$$

Let, 1 mole of A is taken initially and α mole is dissociated at a given T and P.

A remain undissociated $= (1-\alpha)$ moles

B produced $= n\alpha$ mole

Total no. of moles at equilibrium $= 1-\alpha+n\alpha$

$= 1+(n-1)\alpha$

Properties of Gases: Abnormal Vapour Density

Let, V_0 = Volume of the system without any dissociation
V = The actual volume of the vapour due to dissociation

Now, according to the Avogadro's law at constant T and P;

$$\frac{V}{V_0} = \frac{1+(n-1)\alpha}{1} \; ; \quad \text{but} \quad \frac{V}{V_0} = \frac{D_0}{D}$$

$d = \frac{m}{V}$
$m = \text{const.}$
$D_0 V_0 = V D$

So, $$\frac{D_0}{D} = 1+(n-1)\alpha$$

$$\Rightarrow \alpha = \frac{D_0 - D}{(n-1)D} = \frac{M_0 - M}{(n-1)M}$$

D_0 = Expected vapour density from the formula wt. M_0 of the substance

D = Actual vapour density at the given T and P.

M = Determined molecular wt. of the substance by expt.

At high T, $\alpha = 1$ (complete dissociation)

For NH_4Cl and PCl_5, $n = 2$

So $D = \frac{D_0}{2}$. This is confirmed also by expt.

Are there any examples where substances undergo dissociation but do not show abnormal vapour density?

(1) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(2) measurement of density at constant volume also does not lead to abnormal vapour density.

Problem: At 60°c, the density of N_2O_4 gas was found to be 30.2. calculate the percentage of NO_2 molecules by wt. and vol.